

Calc 2 – First-Order ODEs

Worksheet A

1. Solve $y' = -2y$, $y(0) = 5$.
2. Solve the separable ODE $y' = xe^{-y}$.
3. Solve the linear ODE $y' + 2y = 4$.
4. Find the equilibria of $y' = y(1 - y)$ and classify their stability.
5. Apply one step of Euler's method with $h = 0.1$ to $y' = x + y$, $y(0) = 1$.

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Worksheet B

1. Solve $y' = \frac{x}{y}$, $y(0) = 2$.
2. Solve the linear ODE $y' + \frac{1}{x}y = x$, $y(1) = 0$ ($x > 0$).
3. Check that $2xy \, dx + (x^2 + y^2) \, dy = 0$ is exact and solve it.
4. A tank holds 100 L of pure water. Brine with 0.5 kg/L of salt enters at 5 L/min while the well-mixed solution leaves at 5 L/min. Find the salt content $S(t)$.
5. Solve the logistic equation $y' = y(1 - y/4)$, $y(0) = 1$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $y = Ce^{-2x}; y(0) = 5 \Rightarrow \boxed{y = 5e^{-2x}}.$

2. $e^y dy = x dx \Rightarrow e^y = \frac{x^2}{2} + C \Rightarrow \boxed{y = \ln\left(\frac{x^2}{2} + C\right)}.$

3. $\mu = e^{2x}, (e^{2x}y)' = 4e^{2x} \Rightarrow e^{2x}y = 2e^{2x} + C \Rightarrow \boxed{y = 2 + Ce^{-2x}}.$

4. Equilibria $y = 0$ and $y = 1$. $f(y) = y - y^2$, $f'(y) = 1 - 2y$. $f'(0) = 1 > 0$ (**unstable**); $f'(1) = -1 < 0$ (**stable**).

5. $y_1 = y_0 + h(x_0 + y_0) = 1 + 0.1(0 + 1) = \boxed{1.1}.$

Worksheet B.

1. Separate: $y dy = x dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C \Rightarrow y^2 = x^2 + C'$. $y(0) = 2 \Rightarrow C' = 4$. $\boxed{y = \sqrt{x^2 + 4}}.$

2. $\mu = e^{\int \frac{1}{x} dx} = x$. $(xy)' = x^2 \Rightarrow xy = \frac{x^3}{3} + C \Rightarrow y = \frac{x^2}{3} + \frac{C}{x}$. $y(1) = 0 \Rightarrow C = -1/3$.
 $\boxed{y = \frac{x^2}{3} - \frac{1}{3x}}.$

3. $M = 2xy$, $N = x^2 + y^2$. $M_y = 2x = N_x$, exact. Integrate $F_x = 2xy \Rightarrow F = x^2y + g(y)$. Then $F_y = x^2 + g'(y) = x^2 + y^2 \Rightarrow g(y) = y^3/3$. Solution: $\boxed{x^2y + \frac{y^3}{3} = C}.$

4. $S' = \frac{1}{2} \cdot 5 - \frac{S}{100} \cdot 5 = 2.5 - 0.05S$. Linear: $\mu = e^{0.05t}$, $S = 50 + Ce^{-0.05t}$. $S(0) = 0 \Rightarrow C = -50$.
 $\boxed{S(t) = 50(1 - e^{-0.05t})}$ kg.

5. Standard logistic with $M = 4$, $k = 1$: $y = \frac{4}{1+ Ae^{-t}}$. $y(0) = 1 \Rightarrow 1 = \frac{4}{1+A} \Rightarrow A = 3$. $\boxed{y = \frac{4}{1+3e^{-t}}}.$